

Incentives, Collective Action, and Adoption

in Generative Relational Academic Infrastructure

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Abstract

An infrastructure that records how understanding changes faces a set of difficulties that are economic and institutional in place of technical. This paper states them and assesses what can be done about each. It locates the commons dilemma precisely: knowledge is non-rival, so the tragedy that governs pastures has no object there, and the subtractable goods are review capacity, maintainer labour, the instrument, and community trust. It establishes that recognition is a positional good, so competition for standing cannot be dissolved by any degree of openness, and that an open selection process is an all-pay contest in which aggregate effort is dissipated. It proposes one test distinguishing a generative arrangement from a competitive one, namely whether content from unselected lines of work is absorbed with attribution, and shows that the test is also the condition under which a staged selection design survives backward induction. It then treats adoption: the cost of recording falls hardest on those with least standing, the benefits are collective and deferred, and a group adopting the arrangement is selected against in an environment that rewards output rate. Two conditions follow, that the minimal tier be worth adopting by a single participant and that a release emit the artefact the incumbent system accepts, and the empirical record of research infrastructure is read against them. The paper closes with predictions and the conditions under which the account would be shown wrong.

Keywords. collective action; knowledge commons; positional goods; research infrastructure adoption; science policy

Draft. This is a working draft. It is circulated for comment, and its arguments, formulations, and open questions are subject to revision. Correction and criticism are welcome at huangwanhong@serendip.ngo.

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Use of generative artificial intelligence. Generative AI assistants (Anthropic’s Claude and OpenAI’s ChatGPT) were used extensively in preparing this manuscript: for literature survey, for drafting prose from the author’s specifications, for adversarial review of the argument, for the construction of the figures, for consistency and citation auditing, and for typesetting. Formulations and objections arose in the course of that exchange as well as from the author. Every claim, argument, and citation was reviewed and decided by the author, who is responsible for the content and for any errors. Bibliographic details are to be verified against the primary sources before reliance.

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1 Introduction

This section states the question the paper treats, the results it inherits, the position it takes on what can and cannot be secured by design, and the structure of the argument. Its role is to fix the undertaking before any analysis begins. Its method is expository.

A companion paper argues that the record of how an understanding changed has become worth keeping, and a second specifies a protocol for keeping it. Neither addresses the difficulty that determines whether any of it occurs, which is economic and institutional in character.

Recording costs something, and the cost falls on participants. The benefits are partly collective, appear partly at a scale no early adopter reaches, and partly accrue to parties other than the one bearing the cost. A group that adopts such an arrangement operates in an environment that rewards output rate and rewards nothing else, so the adoption is a decision taken against the local gradient. And every mechanism the protocol supplies for making contributions visible operates within a system whose central good, recognition, is positional and therefore fixed in aggregate.

This paper treats those difficulties. It does not argue that they are surmountable. It states each precisely, distinguishes those a design can address from those it cannot, and specifies what would have to be true for adoption to occur.

Four results from the companion papers are used without re-argument.

The evidential function of a published artefact rests on a separating condition, and a uniform fall in production cost weakens it while leaving the artefact in place.

A record of a process is as cheaply fabricated as the artefact whose function it would inherit, unless its credibility follows from distribution across parties who did not coordinate.

Systems representing scholarly reasoning have existed since 1970 and were not adopted, and the recorded cause is that the work of recording fell on the party who gained least from it.

And the protocol specified in the companion paper defines three conformance tiers, the lowest of which requires three acts and one participant, together with a release that emits a document of the kind the incumbent reward system accepts.

The paper takes a position on what an infrastructure can secure, and states it here so that the later sections are read correctly.

A protocol can alter the distribution of costs and can make certain acts cheap. It cannot create an incentive where none exists, cannot dissolve competition for a positional good, and cannot supply a norm that a community has not adopted. Where the analysis reaches one of these limits, the paper says so and stops, and the section on falsification names what would show even the modest claims wrong.

The paper also declines a rhetorical move common in this literature. It does not argue that the present arrangement is so defective that any alternative is worth trying. The present arrangement performs two functions this account must preserve, concentration of resources on evidenced capacity and the reward that makes early disclosure rational, and a proposal that degraded either without replacement would leave the system worse.

The paper proceeds in four parts.

The first locates the commons dilemma. Knowledge is non-rival, so the appropriation problem has no object there, and the subtractable goods are identified and treated separately: review capacity, maintainer labour, the instrument, and community trust. Recognition is then shown to be positional, which bounds every claim made later.

The second treats selection. An open competition is an all-pay contest with dissipated aggregate effort, and the question is what distinguishes an arrangement that generates from one that merely selects. The absorption test is proposed, and staged selection designs are examined under backward induction.

The third treats adoption: the distribution of overhead, the selection pressure on adopting groups, network externalities against the tier structure, compatibility with the incumbent reward system, and what the record of research infrastructure shows.

The fourth states predictions and the conditions under which the account would be shown wrong.

2 Scope, Terms, and the Method of Assessment

This section fixes the terms the analysis uses and the standard by which its claims are to be judged. Its role is to prevent two failures common in writing about scholarly infrastructure: the use of economic vocabulary without its conditions, and the presentation of a mechanism as evidence that the mechanism works. Its structure treats the goods at issue, the terms, the evidential standard, and what the paper counts as a solved difficulty. Its method is stipulative.

2.1 The Goods at Issue

Definition 2.1. (Rivalry and excludability). *A good is rival where one party's use diminishes what is available to another, and excludable where parties can be prevented from using it. A public good is neither; a common-pool resource is rival and hard to exclude from; a club good is excludable and non-rival within the club.*

The distinctions are standard and are stated because the argument of the next section turns on applying them carefully to goods that are usually treated together.

Definition 2.2. (Positional good). *A good is positional where its value to a holder consists in the holder's rank among others, so that the aggregate quantity available is fixed by construction and increases in supply do not increase it (Hirsch, 1976).*

2.2 Terms

Adoption means that a party begins to conduct work under the protocol at a declared tier, and continues to do so. A party who records one trajectory and stops has not adopted.

Overhead means the effort a participant expends that would not have been expended without the arrangement, measured against the participant's own purposes and not against an ideal.

Private benefit means a benefit accruing to the party bearing a cost, at the scale at which that party operates, without the participation of others.

Generative arrangement and *competitive arrangement* are defined in the section on the absorption test, since the distinction between them is a result of the paper and not a stipulation.

2.3 The Evidential Standard

Requirement 2.3. (Marking of the evidential status of claims). *Every claim in this paper is marked as one of three kinds. A formal claim follows from stated assumptions and is established by argument. An empirical claim is supported by cited studies, with the population and the strength of the support stated. A conjecture is neither, and is marked as such wherever it appears.*

The instrument matters here more than in the companion papers, because the literatures this paper draws on differ greatly in strength. The economics of positional goods and of contests is formal. The evidence on assessment reliability and on the cost of competition is empirical and reasonably strong. The evidence on what makes research infrastructure adopted is thin, largely retrospective, and drawn from cases selected because they are known.

2.4 The Standard for Treating a Difficulty as Addressed

Requirement 2.4. (Standard for a solution). *A difficulty is treated as addressed only where the paper states a mechanism, identifies the party who bears its cost, and names the condition under which the mechanism fails. A difficulty for which a mechanism is described without an identified cost-bearer is recorded as unaddressed.*

The standard is stricter than the literature's ordinary practice and is adopted because the failure it guards against is the characteristic failure of this field. Proposals for scholarly infrastructure routinely describe arrangements in which some party performs work that benefits others, without stating why that party would perform it, and the systems built on such proposals are the ones whose non-adoption the companion paper documents.

2.5 Summary of Results

This section has fixed the goods at issue through Definition 2.1 and 2.2, has fixed adoption, overhead and private benefit, and has deferred the definition of generative and competitive arrangements to the section that derives it. Requirement 2.3 requires every claim to be marked formal, empirical or conjectural, and records that the literatures relied on differ greatly in strength, with the evidence on infrastructure adoption the weakest. Requirement 2.4 fixes the standard for treating a difficulty as addressed: a mechanism, an identified cost-bearer, and a stated condition of failure.

3 Non-Rivalry, Subtractability, and the Location of the Commons Dilemma

This section determines where in the arrangement a commons dilemma actually arises. Its role is corrective: writing about knowledge commons routinely applies the tragedy of the commons to knowledge itself, and the application is a category error whose consequence is that the real dilemmas go untreated. Its structure treats the character of knowledge as a good, the failure that does attach to it, the four subtractable goods in the arrangement, and the inverse risk the design itself creates. Its method is application of the distinctions fixed in Section 2, with each claim marked.

3.1 The Character of Knowledge as a Good

Knowledge is non-rival. A party's use of a result does not diminish what remains available, and this is true of a trajectory record as it is of a published claim. It is also imperfectly excludable once disclosed, which is the observation from which the economics of research funding proceeds (Arrow, 1962).

Claim 3.1. *The tragedy of the commons has no object in knowledge. That analysis concerns a rival resource degraded by uncoordinated appropriation, and no quantity of reading, citing, absorbing or forking diminishes what a trajectory record holds. Arguments transferring the tragedy to knowledge commons transfer a mechanism whose precondition is absent. This claim is formal.*

The point is not pedantic. A commons dilemma misdiagnosed as appropriation invites remedies of the appropriation kind, which are restrictions on use, and the companion paper on restriction shows that restrictions are justified only on stated grounds none of which is overuse.

3.2 The Failure That Does Attach

What knowledge suffers is the opposite failure.

Claim 3.2. *Knowledge is a public good, so the failure predicted for it is underprovision and not overuse: parties bear the cost of producing what others consume, and produce less than the aggregate would justify. The relevant analysis is that of collective action under diffuse benefit (Olson, 1965), and its predictions differ from those of the appropriation analysis at almost every point. This claim is formal.*

Two consequences follow for this series. Remedies that increase the cost of using knowledge address a failure that does not obtain and worsen the one that does. And the design's central difficulty, that recording benefits parties other than the recorder, is an instance of the underprovision problem rather than a novel feature of the proposal.

3.3 The Four Subtractable Goods

The dilemma does arise, and it arises on the goods the arrangement consumes in order to produce knowledge.

Review and assessment capacity is the first and the most consequential. Reading a proposed transition carefully, judging a challenge, or assessing a stranger's contribution takes the time of a party competent to do it, and that time is rival, limited, and unpaid. An arrangement that lowers the cost of contributing while leaving the cost of assessing unchanged increases demand on a fixed supply, which is a congestion problem in its standard form.

Maintainer labour is the second. Someone keeps the record coherent, resolves duplicate states, curates the register of open problems, and answers newcomers. The open-source record documents both the concentration of this labour on very few parties and the consequences when they stop (Eghbal, 2016). This is empirical, drawn from a documented and much-cited survey of digital infrastructure.

The instrument is the third, where the work requires one, and it is rival in the ordinary sense. It is treated in the companion paper on restriction and is named here for completeness.

Community trust is the fourth and the least tangible. A community's willingness to extend good faith to unknown contributors is consumed by bad experiences and replenished slowly, and an open entry path draws on it directly.

Claim 3.3. *The commons dilemma in this arrangement attaches to review capacity, maintainer labour, the instrument and community trust, and not to knowledge. Of these, review capacity is the one the design most directly strains, since the protocol lowers the cost of contributing and lowers the cost of assessing by nothing. This claim is formal, and the empirical support for the second sentence is absent because the arrangement has not been operated.*

Ostrom's design principles for the governance of common-pool resources apply to these four goods and not to the knowledge produced from them (Ostrom, 1990), which is a distinction the knowledge-commons literature blurs and which determines where governance effort should be spent.

3.4 Rationing without a Ranking

The congestion problem admits an obvious remedy that this series has excluded, and the exclusion has a cost that must be stated.

Congestion of assessment is ordinarily managed by ranking: attention is directed to contributions with high scores, and the rest go unread. The companion papers exclude every scalar quantity over participants and trajectories, so this remedy is unavailable within the protocol.

Three alternatives exist and none is supplied by the protocol.

Reciprocal obligation, under which a party receives assessment in proportion to the assessment they perform, is the standard proposal for peer review and has no deployed instance at scale.

Bounded intake, under which a community accepts a fixed number of open contributions in a period, rations without ordering and displaces the difficulty onto whoever administers the bound.

And triage by a maintainer, which is what open-source projects do, and which rein-

troduces the position the companion paper's admissibility condition governs.

Claim 3.4. *The exclusion of scalar measures leaves the congestion of assessment capacity unaddressed at the protocol level, and the remedies available operate at the level of a community's charter. This is recorded as an unaddressed difficulty under the standard of Requirement 2.4, since no mechanism is stated with an identified cost-bearer.*

3.5 The Inverse Risk

One further hazard belongs here because it is created by a mechanism this series proposes rather than inherited from the situation.

Attribution of absorbed content, multiplied across many small contributions, resembles the fragmentation documented for biomedical patents, in which downstream work requires so many permissions that it does not occur (Heller and Eisenberg, 1998). The companion specification answers this by excluding any veto: an absorption link confers attribution and confers no power to prevent, condition or reverse use.

Claim 3.5. *The anticommons risk created by fine-grained attribution is answered by separating attribution from exclusion, which is the moral-rights construction. The answer is complete with respect to legal fragmentation and incomplete with respect to social fragmentation, since a community may treat attribution claims as though they carried veto power whatever the specification says. The second sentence is a conjecture.*

3.6 Summary of Results

This section has located the dilemma. Knowledge is non-rival, so the tragedy of the commons has no object there, and arguments transferring it import a mechanism whose precondition is absent. The failure that does attach to knowledge is underprovision, whose remedies differ from those of appropriation at almost every point. The dilemma arises instead on four subtractable goods: review capacity, maintainer labour, the instrument, and community trust, with Ostrom's design principles applying to these and not to the knowledge produced from them. Review capacity is the good the design most directly strains, since contributing is made cheaper and assessing is not. The exclusion of scalar measures removes the ordinary remedy of ranking, and the three alternatives available operate at the charter level, so the congestion of assessment is recorded as an unaddressed difficulty. The anticommons risk created by fine-grained attribution is answered legally by excluding veto rights and is not answered socially.

4 Review Capacity and the Provision of a Congestible Good

This section treats the good the design most directly strains. Its role is to state the congestion problem precisely, to assess the remedies available, and to record what remains unaddressed. Its structure treats the asymmetry the protocol introduces, the present provision of assessment, the three remedies, the position the companion papers take, and the residual difficulty. Its method is analysis with each claim marked, and the section reaches no solution.

4.1 The Asymmetry Introduced

Every mechanism in the specified protocol lowers the cost of contributing. A challenge references an identified state and is registered in an affirmative action; a connection from a stranger requires no prior standing; the register of unresolved states publishes the points at which contribution is invited.

No mechanism lowers the cost of assessing. Judging whether a challenge is sound, whether a connection is apt, or whether a stranger's verification was competently performed requires the attention of a party who understands the material, and the protocol supplies nothing that shortens the judgement.

Claim 4.1. *The protocol alters the ratio between the cost of contributing and the cost of assessing, in the direction that increases demand on assessment. Where such an arrangement attracts participation, congestion of assessment capacity follows, and the congestion is a consequence of the design working rather than of its failing. This claim is formal.*

4.2 The Present Provision of Assessment

Assessment in the incumbent arrangement is provided by unpaid reviewers under a norm of reciprocity that is weakly enforced and weakly monitored, and the provision is under strain independently of anything proposed here. The empirical literature reports low agreement between reviewers on the same material (Cole et al., 1981; Pier et al., 2018), and analyses of competition report that the aggregate effort expended in preparing and assessing proposals is large relative to what is allocated (Gross and Bergstrom, 2019).

Claim 4.2. *The assessment capacity available to scholarship is fixed in the short run, is unpaid, is provided under a norm rather than a contract, and is already strained. A design increasing demand on it must either supply new capacity or ration what exists. This claim is empirical for the first sentence, with the cited studies concerning grant and journal review, and formal for the second.*

4.3 The Three Remedies and Their Costs

Ranking directs attention to the contributions scored highest and leaves the rest unread. It is the remedy in general use, it is effective, and it is excluded by the no-scalar requirement of the companion papers, for reasons developed there: a measure adopted to direct attention becomes the object of effort, and the categories it recognises become the categories the field maintains (Espeland and Sauder, 2007).

Reciprocal obligation conditions the assessment a party receives on the assessment they perform. It is proposed regularly for peer review, it addresses the free-riding directly, and it has no instance at scale. Two difficulties are known: assessment quality is unobservable, so a party may discharge an obligation cheaply; and the parties whose contributions most need assessment are often those least able to supply it, so the mechanism excludes newcomers in the manner the entry-path requirement forbids.

Bounded intake accepts a fixed number of open contributions in a period and closes. It rations without ordering, so it is compatible with the no-scalar requirement, and it displaces the difficulty onto whoever sets and administers the bound. It also wastes: a contribution arriving after the bound is not assessed regardless of its merit, and the waste falls arbitrarily.

Triage by a maintainer is what open-source projects do in practice. A small number of parties decide what receives attention, and the arrangement is efficient and reintroduces exactly the position the companion paper's admissibility condition governs, with the record of that paper's evidence on concentration (Shaw and Hill, 2014).

4.4 The Position the Companion Papers Take

The protocol supplies none of these and permits three of them, which is a position worth stating plainly rather than presenting as neutrality.

Ranking is forbidden at the protocol level and is available at the application level to any party willing to build it, and the companion paper on failure modes records that a widely adopted external ranking would reintroduce the dynamic the exclusion exists to prevent.

Reciprocal obligation, bounded intake and maintainer triage are all charter matters. The protocol records who assessed what and when, which makes a reciprocity norm auditable by the community that adopts it, and supplies nothing further.

Claim 4.3. *The protocol makes assessment activity visible and does not provide for assessment capacity. Visibility is a precondition for a community to operate a reciprocity norm and is not itself such a norm. This claim is formal.*

4.5 The Residual Difficulty

Under the standard fixed in Section 2, the difficulty is recorded as unaddressed, and the record should be precise about what is missing.

A mechanism exists in each of the three permitted remedies. What is absent in each is an identified party who bears the cost under conditions that make bearing it rational. In reciprocal obligation the cost falls on the party performing assessment and the return is a claim on future assessment whose value depends on others honouring the same norm. In bounded intake the cost falls on the administrator of the bound. In triage it falls on the maintainer, and the open-source record is that this party burns out (Eghbal, 2016).

Claim 4.4. *Congestion of assessment capacity is the difficulty in this series least addressed by the design and most likely to determine whether an arrangement conforming to the protocol survives at scale. Nothing in the specification bears on it, and the remedies available to a community all place an unfunded burden on a small number of parties. This claim is a conjecture as to the first sentence and formal as to the second.*

4.6 Summary of Results

This section has treated assessment capacity. The protocol lowers the cost of contributing and lowers the cost of assessing by nothing, so congestion follows from the design working. Present provision is unpaid, norm-governed, already strained, and documented as unreliable and costly. Four remedies are assessed: ranking, which is effective and excluded; reciprocal obligation, which has no instance at scale and excludes newcomers; bounded intake, which rations without ordering and wastes arbitrarily; and maintainer triage, which is efficient and reintroduces the position the admissibility condition governs. The protocol permits the last three, supplies none, and contributes visibility of assessment activity, which is a precondition for a reciprocity norm and not a norm. The difficulty is recorded as unaddressed, since each remedy lacks an identified cost-bearer for whom bearing the cost is rational.

5 Maintainer Labour and the Underprovision of Infrastructure

This section treats the second subtractable good. Its role is to establish what maintenance an arrangement of this kind requires, who performs it under present conditions, and what the record shows about the sustainability of that arrangement. Its structure treats the work itself, the evidence from software infrastructure, the disanalogy with software, the funding question, and what the protocol does about it. Its method is analysis against the empirical record, with the strength of that record stated.

5.1 The Work

An arrangement conforming to the protocol requires continuing work that is not research.

Registration of other parties' transitions, which the specification makes cheap and does not make free. Resolution of duplicate or near-duplicate states. Curation of the register of unresolved states, so that it remains a list of open problems in place of a list of everything nobody has closed. Response to strangers arriving through the entry path. Maintenance of the charter, its disclosure classes and its membership. Custody of the record, its deposit and its succession arrangements.

Claim 5.1. *The maintenance an arrangement requires grows with its adoption, and the parties who benefit from maintenance are not the parties who perform it. This is the structure of the disparity documented for collaborative systems generally (Grudin, 1988), applied to the operating rather than the recording side. This claim is formal.*

5.2 The Evidence from Software Infrastructure

The nearest evidence concerns open-source software, and it is worth stating carefully because it is usually cited as an encouragement and reads as a warning.

The survey record documents that widely used software infrastructure is maintained by very few parties, often unpaid, that the maintenance burden rises with adoption while the contributor base does not, and that under-maintenance of critical components is com-

mon (Eghbal, 2016). The concentration of decision rights in open collaborative communities is documented separately and independently (Shaw and Hill, 2014).

Claim 5.2. *Open-source software is not evidence that voluntary maintenance of shared infrastructure is sustainable. It is evidence that such maintenance concentrates on very few parties, that those parties are frequently unpaid, and that the arrangement fails silently until something breaks. This claim is empirical, resting on survey and case evidence rather than on a systematic study of a defined population.*

5.3 The Disanalogy

Two differences make the software case an imperfect guide, and they run in opposite directions.

Software has a verification mechanism the arrangement here lacks: a contribution either passes the tests or does not, so a maintainer's assessment cost per contribution is low. Assessment of a challenge to a claim has no such shortcut, so maintenance here is more expensive per unit than in the software case.

Software has a user population far larger than its contributor population, and a corresponding funding argument: firms whose products depend on a component have an interest in its maintenance and sometimes act on it. A trajectory record has no comparable population of dependent commercial users, so the funding argument that supports open-source maintenance does not transfer.

5.4 The Funding Question

Claim 5.3. *No party in the present arrangement has an interest in funding the maintenance of a trajectory commons that is proportional to the benefit it would receive. Institutions gain evidence of contribution, funders gain selection information, and unaffiliated scholars gain visibility, and each of these benefits is diffuse while the maintenance cost is concentrated, which is the condition under which collective action fails (Olson, 1965). This claim is formal, and the identification of the three beneficiaries is a conjecture.*

Olson's analysis names one exception that is worth recording, since it is the mechanism by which comparable infrastructure has in fact been funded. Where a single party's benefit exceeds the whole cost of provision, that party provides the good alone and others free-ride. In scholarly infrastructure this describes the funder or the institution that decides an arrangement serves its own purposes and pays for it, and it also describes the failure mode in which the paying party's purposes become the arrangement's purposes.

5.5 The Protocol's Contribution to Maintenance

Three things, none of which is a solution.

It makes maintenance work visible. Registrations, administrative operations, class changes and charter amendments are all recorded with their performers, so a community can see who is carrying the arrangement. Visibility is a precondition for compensation and is not compensation.

It bounds the maintainer's authority through portability, which the companion paper establishes, and which has a cost here: a maintainer whose authority is bounded and whose labour is unpaid holds a position with the burdens of authority and few of its returns.

And it permits the work to be distributed, since registration may be performed by any participant and the specification requires no single maintaining party. Whether it is distributed in practice is a fact about communities.

Claim 5.4. *Maintenance of an arrangement conforming to the protocol is unfunded, concentrates on few parties, grows with adoption, and is recorded as an unaddressed difficulty under the standard of Section 2. This claim is formal as to the first three properties, given the argument above, and the record of comparable arrangements supports them empirically.*

5.6 Summary of Results

This section has treated maintainer labour. The work required is registration, duplicate resolution, curation of the open register, response to entrants, charter maintenance and custody, and it grows with adoption while benefiting parties other than those who perform it. The software record documents concentration on few and often unpaid parties, rising burden, and silent under-maintenance, and is a warning rather than an encouragement. Two disanalogies are recorded: assessment here lacks the shortcut that tests supply, and no population of dependent commercial users exists to fund the work. No party's interest is proportional to the benefit it would receive, which is the standard condition of collective-action failure, with Olson's privileged-provider exception naming both how such infrastructure gets funded and how its purposes are captured. The protocol makes maintenance visible, bounds the maintainer's authority at a cost to the maintainer, and permits distribution of the work, and the difficulty is recorded as unaddressed.

6 The Positional Character of Recognition and the Limits of Openness

This section fixes the boundary that constrains every claim the series makes about incentives. Its role is to establish that competition for standing cannot be dissolved by any property of an infrastructure, and to determine what may be changed instead. Its structure treats the positional character of recognition, the consequences for the proposal, the two things that can be altered, and the objection that the argument proves too much. Its method is application of a standard economic distinction, with the empirical residue marked.

6.1 Recognition as a Positional Good

Standing in a scholarly field consists in a party's rank among others: appointment, promotion, invitation, citation and reputation are all allocated relative to a comparison set. A good of this kind is positional in the sense fixed in Section 2, and the aggregate available is fixed by construction (Hirsch, 1976).

Claim 6.1. *The aggregate quantity of standing available in a field is fixed independently of how much work is produced, how visible contributions are made, or what infrastructure records them. Any arrangement that makes more contributions visible redistributes standing and does not increase it. This claim is formal.*

The cumulative-advantage literature bears on how the fixed quantity is distributed rather than on whether it is fixed (Merton, 1968), and it establishes that the distribution is unequal in a way that compounds. Both results hold together: the quantity is fixed and its distribution favours those who already hold it.

6.2 The Consequences for the Proposal

Three follow directly, and each removes a claim the series might otherwise be read as making.

Attribution of previously invisible contributions does not create recognition for those who receive it, except at the expense of parties who presently hold it. The companion papers' attribution mechanisms are therefore redistributive, and a party who presently benefits from the invisibility of others' contributions is made worse off by them. This is a reason to expect resistance and is not an argument against the mechanism.

An arrangement cannot make competition for standing less intense by being open. Openness alters who may enter the competition and alters nothing about what is competed for.

And any claim that the arrangement makes scholarship less competitive should be struck wherever it appears in the series. What may be claimed is that it changes what competition rewards.

Claim 6.2. *The series may claim that the arrangement changes what parties compete for. It may not claim that the arrangement reduces competition, dissolves hierarchy, or increases the recognition available to those who presently lack it, and any passage implying otherwise is an error to be corrected. This claim is formal.*

6.3 The Dimensions Open to Alteration

Two things, and stating them precisely is what the section is for.

The basis of comparison. A field allocating standing on artefact counts and a field allocating it on recorded contribution to changes in understanding are both competitive and are competitive over different things. The companion paper's argument is that the first basis has weakened as evidence, so a change of basis is available in a way it was not before. Whether a community makes the change is a question about the community.

The visibility of the contribution. A party whose contribution altered another's work presently has no way of showing that it did. The attribution mechanism supplies one. The party's standing rises only if the community treats such contributions as counting, which returns to the first item.

Claim 6.3. *The arrangement supplies a record on which an alternative basis of comparison could operate. It does not supply the recognition, which is conferred by communities, and it*

does not supply the norm under which recorded contribution counts. This claim is formal, and whether communities adopt such a basis is a conjecture on which the series offers no evidence.

6.4 The Objection That the Argument Proves Too Much

An objection deserves treatment because it applies to the whole series.

If competition for a fixed quantity is unaffected by infrastructure, then no infrastructure matters, and the objection concludes that the series is arguing for something whose value it has just denied.

The objection fails, and the reason locates the value of the proposal precisely. What is fixed is the aggregate of standing. What is not fixed is which activities produce it, who is eligible to compete, what evidence is admissible, and what is lost when work is not recorded. The reuse of abandoned material, the redirection of attention across fields, the interpretability of a decision years later: none of these concerns standing, and none is bounded by the positional argument.

Claim 6.4. *The positional argument bounds the claims the series may make about recognition and bounds nothing it claims about the retention of work, the reuse of abandoned material, or the interpretability of a record. The two sets of claims should be kept apart wherever they appear. This claim is formal.*

6.5 Summary of Results

This section has fixed the boundary. Recognition is a positional good, so the aggregate available is fixed and any arrangement making contributions visible redistributes it in place of increasing it, with cumulative advantage governing how the fixed quantity is distributed. Three consequences follow: attribution mechanisms are redistributive and will be resisted by those who benefit from present invisibility; openness alters who competes and not what is competed for; and no claim that the arrangement reduces competition may stand anywhere in the series. Two things can be altered: the basis on which standing is allocated, which is available because the artefact has weakened as evidence, and the visibility of contributions, whose effect on standing depends on the first. The objection that the argument proves too much fails, since the positional bound applies to recognition alone and leaves untouched every claim about retention, reuse and interpretability.

7 Rent Dissipation and the Character of an Open Selection

This section examines what happens when an arrangement opens a selection to many parties. Its role is to establish that openness of a competition is not a benefit in itself, and to prepare the distinction the following section draws. Its structure treats the contest structure, the dissipation result, its application to open selection in research, the two responses commonly offered, and what remains. Its method is the application of contest theory, with the transfer's boundaries stated.

7.1 The Contest Structure

An arrangement in which many parties prepare proposals, one is selected, and the others receive nothing has a standard form. Effort is expended by every entrant, the effort is sunk whether or not the entrant wins, and the prize is awarded to one.

Definition 7.1. (Dissipation). *Dissipation is the aggregate effort expended by entrants in a contest, in excess of what the selection required. Where entrants are numerous and the prize is valuable, dissipation may approach or exceed the value of the prize.*

The result is standard in contest theory and has been applied to scientific funding directly, where analyses report that the aggregate preparation effort across applicants is large relative to the funds distributed (Gross and Bergstrom, 2019). The transfer to the present case carries two boundaries: the models assume entrants who choose effort with knowledge of the prize and of the number of competitors, and neither assumption holds cleanly in research.

7.2 The Application to Open Selection

An open selection in research has a particular feature that worsens the arithmetic.

Claim 7.2. *Opening a selection to more entrants increases aggregate dissipation, other things equal. Where the selection is opened in the name of fairness, the fairness is obtained at a cost in aggregate effort that falls on the entrants, and disproportionately on those for whom the preparation effort is a larger share of their available time. This claim is formal, given the contest structure.*

The observation bears directly on a design the companion papers might otherwise have endorsed. An arrangement in which many parties develop proposals in the open, one is chosen for the instrument or the funding, and the rest are discarded, is a more inclusive contest and not a different kind of thing. It is the incumbent arrangement with a wider entrance and higher aggregate cost.

7.3 The Two Responses Commonly Offered

The learning response holds that unsuccessful entrants gain from the preparation, so the effort is not wasted. This is true in part and does not close the gap: the gain accrues to the entrant privately, is unrecorded, and is lost when the entrant moves on, which is the condition the companion papers exist to address.

The signalling response holds that the effort produces information about the entrants that the selector uses. This is also true in part, and the evidence on assessment reliability limits how much information is in fact extracted (Cole et al., 1981; Pier et al., 2018).

Claim 7.3. *Neither response defeats the dissipation result. The learning response identifies a private and unrecorded gain, and the signalling response identifies an informational gain whose magnitude the evidence on assessment reliability bounds. This claim is empirical as to the second, resting on studies of grant review, and formal as to the structure.*

7.4 The Question Passed to the Following Section

The section reaches a position that the following one develops.

Selection is unavoidable where a resource is scarce, as the companion paper on restriction establishes. Opening the selection does not alter its character. What can be altered is what becomes of the effort expended by the entrants who are not selected, and the learning response points at the answer without reaching it: the gain is private and unrecorded, and an arrangement that recorded it would convert a dissipated effort into a retained one.

Claim 7.4. *The difference between an arrangement that dissipates the effort of unselected entrants and one that retains it lies in whether the content of unselected work enters the work that proceeds, with attribution. This is a claim about a mechanism and is developed in the following section. This claim is formal.*

7.5 Summary of Results

This section has established that an open selection is a contest and that opening it increases aggregate dissipation, with the burden falling disproportionately on entrants for whom preparation is a larger share of available time. The learning and signalling responses each identify a partial offset and neither defeats the result, the first because the gain is private and unrecorded, the second because the evidence on assessment reliability bounds the information extracted. Selection is unavoidable under scarcity and openness does not change its character. What can be altered is whether the content of unselected work enters what proceeds, which is developed next.

8 The Absorption Test and the Distinction between Generative and Competitive Arrangements

This section states the distinction the paper's title implies and gives it an observable form. Its role is to supply a test that separates an arrangement that generates from one that merely selects, and to state what the test does and does not establish. Its structure treats the definitions, the test, its checkability, its relation to the criteria of the companion papers, and its limits. Its method is definition followed by claim, with the observable stated so that a reader can apply it to an existing arrangement.

8.1 The Definitions

Definition 8.1. (Competitive and generative arrangements). *An arrangement is competitive where the work of entrants who are not selected leaves no trace in the work that proceeds. An arrangement is generative where content originating in unselected work enters the work that proceeds, and is attributed to the party who produced it.*

The definitions turn on what becomes of unselected work and not on the openness of entry, the number of participants, or the manner of the decision. An open arrangement in which losing proposals are discarded is competitive under this definition, and

a closed arrangement in which a rejected proposal's method is taken up and credited is generative.

8.2 The Test

Requirement 8.2. (The absorption test). *An arrangement is assessed by examining the record of the work that proceeded and counting the transitions whose content originated in work that was not selected, with attribution to the originating party. Where such transitions are absent, the arrangement is competitive whatever its constitution states.*

Three properties make the test usable.

It is checkable from the record the companion specification already requires. Absorption links carry the identifier of the state absorbed from and the party who produced it, so the count requires no new instrument.

It is falsifiable in a way that is uncomfortable for the proposal. Any arrangement, including one conforming to the protocol, may be examined and found to absorb nothing.

And it applies to existing arrangements without the protocol. A funding programme, a laboratory, or a journal may be examined for whether the work that proceeded took up content from what it declined, though in the absence of a record the examination is laborious and its result uncertain.

8.3 Relation to the Criteria of the Companion Papers

The test is not a fourth criterion added to the series. It is the criterion the companion papers state, applied at the point where a selection occurs.

Those papers hold that value which returns to the relation that produced it is generative, and value which accrues to an attractor outside that relation is not. In a competitive arrangement the effort of unselected entrants accrues to the selecting party, who obtains information, and to the selected entrant, who obtains the resource, and returns nothing to the relations that produced it. In a generative arrangement part of it returns as attributed content.

Claim 8.3. *The absorption test is the folding criterion of the companion papers applied to a selection, and the series therefore employs one criterion at four positions rather than four criteria. This claim is formal.*

8.4 The Limits of the Test

Four, and the third is the one that prevents the test from becoming a metric.

Absorption is necessary and not sufficient for an arrangement to be worth having. An arrangement may absorb diligently and produce nothing of value, and the test bears on the disposition of effort and not on the quality of the result.

Absorption may be performed badly. Content may be taken up in a way that misrepresents what the originating party held, and the attribution then records a debt that the

originator would not recognise. The specification's provision for contested attribution is what such a party has.

Counting is forbidden. An absorption count over trajectories would be a scalar of the kind the series excludes, and it would be farmed within a season by parties recording nominal absorptions. The test is applied by a reader examining a particular arrangement and reaching a judgement, and it is not a quantity the protocol computes or an implementation displays.

And the test says nothing about generativity in the sense of capacity. Two arrangements may absorb identically while the relations within them differ entirely, which is the general limit established in the companion papers and which applies here as everywhere.

Claim 8.4. *The absorption test distinguishes what becomes of unselected effort and does not measure the quality, the quantity or the generative capacity of what an arrangement produces. Its use is diagnostic and comparative, applied by readers to particular arrangements, and any implementation computing an absorption score has violated the exclusion of scalar measures. This claim is formal.*

8.5 Summary of Results

This section has fixed the distinction. Definition 8.1 makes an arrangement competitive where unselected work leaves no trace and generative where content from unselected work enters what proceeds with attribution, turning on the disposition of effort in place of the openness of entry. Requirement 8.2 states the test, which is checkable from the absorption links the specification already requires, is falsifiable against arrangements conforming to the protocol, and applies laboriously to arrangements without a record. Claim 8.3 identifies the test as the folding criterion of the companion papers applied to a selection, so the series employs one criterion at four positions. Four limits are fixed: absorption is necessary and not sufficient, may be performed badly, must not be counted, and says nothing about generative capacity.

9 Staged Selection and the Backward Induction of Strategic Behaviour

This section examines the design most naturally proposed for combining open generation with unavoidable selection. Its role is to show that the design fails as ordinarily stated, and to identify the condition under which it does not. Its structure treats the design, the induction, the two behaviours it predicts, the condition that repairs it, and what remains unrepaired. Its method is game-theoretic reasoning of an informal kind, with the reasoning's limits stated.

9.1 The Design

The design has three stages. In the first, participants work in the open with no ranking and no allocation, and diversity is encouraged. In the second, lines of work are developed collectively, with objection and revision. In the third, a selection occurs and resources

are allocated to one or a few of the lines.

Its attraction is evident. The generative benefits are obtained in the first two stages, the unavoidable selection is confined to the third, and the openness of the early stages is protected by the declaration that no ranking occurs there.

9.2 The Induction

Claim 9.1. *Where a later stage allocates a scarce resource and every participant knows it, the earlier stages are not unranked. They are ranked with a delay, and rational participants act in the earlier stages with the later stage in view. A declaration that a stage is non-competitive does not make it non-competitive. This claim is formal, under the assumption that participants anticipate the selection and prefer to be selected.*

The assumption is weak and the conclusion is robust to relaxing it. It is enough that some participants anticipate the selection; their behaviour then imposes the same pressure on the rest.

9.3 The Two Behaviours Predicted

Withholding. A participant with a strong idea has reason to disclose it late, and ideally at the moment when disclosure and selection are close enough that the idea cannot be developed by a better-resourced party first. The first stage therefore fills with the material participants are least concerned about, which is the material least worth having.

Adversarial objection. In the second stage, a participant improving a rival's line of work raises the probability that the rival is selected. The rational act is to object in ways that damage rather than to object in ways that improve, and objection is exactly the mechanism the arrangement relies on for its generative benefit.

Claim 9.2. *A staged design as ordinarily stated predicts that the exploration stage receives weak material and that the development stage produces destructive rather than constructive objection. Both predictions follow from the induction and both concern the mechanisms on which the design's benefits depend. This claim is formal.*

9.4 The Condition That Repairs It

The repair is not a further stage or a stronger declaration. It is the mechanism of the previous section, applied as an incentive rather than as a diagnostic.

Claim 9.3. *Where content absorbed into a selected line is attributed to the party who produced it, and where the attribution is recorded durably and is recognised, contributing to a rival's line carries a positive expected return, and objecting in ways that improve carries a return that objecting in ways that damage does not. Attributive absorption is therefore the condition under which a staged design survives backward induction, rather than a courtesy extended to unselected parties. This claim is formal as to the incentive structure and conditional on the recognition of attribution, which is a fact about communities.*

Two features of the repair are worth noting.

It operates on the second stage more strongly than on the first. An objection that improves a rival's line is recorded and attributed immediately, while a strong idea disclosed early is protected only by registration, which the companion papers supply and which converts exposure into a dated claim without generating anything.

And it requires the attribution to be worth something, which returns the argument to the positional analysis. Attribution redistributes standing only where a community counts it, and a community that counts only selected outcomes leaves the repair inoperative.

9.5 The Residual Difficulties of the Staged Design

Three items.

Withholding is mitigated and not removed. Registration protects a claim and does not compensate the party for the benefits foregone by not circulating the idea, and a participant who withholds loses the objections that would have improved the work. The arrangement makes withholding costly to the withholder and leaves it available.

The repair depends on a norm the design cannot supply. Where absorbed contributions are recorded and ignored by hiring committees and funders, the expected return is zero and the induction reasserts itself.

And the analysis assumes participants who understand the arrangement. A design whose incentive properties depend on participants reasoning about later stages will behave differently among participants who do not, and the direction of the difference is not obvious.

Claim 9.4. *Staged selection is a workable design only in conjunction with attributive absorption and only where a community recognises absorbed contributions. Absent either, the design predicts weak early disclosure and destructive objection, and is worse than an undisguised competition, since it obtains the appearance of openness while producing the behaviour of a contest. This claim is formal, and the comparison in the second sentence is a conjecture.*

9.6 Summary of Results

This section has examined the staged design. Claim 9.1 establishes that a stage followed by an allocating stage is ranked with a delay, and Claim 9.2 predicts the two consequent behaviours: withholding of the strongest material from the exploration stage, and objection that damages in place of improving during development. Claim 9.3 identifies attributive absorption as the condition under which the design survives the induction, since crediting content taken from unselected lines makes contribution to a rival carry positive expected return; the condition operates more strongly on development than on exploration and depends on communities counting attribution. Three items remain unrepaired: withholding is made costly and not removed, the repair depends on a recognition norm the design cannot supply, and the analysis assumes participants who reason about later stages. Claim 9.4 states the conclusion, that a staged design without attributive absorption is worse than an undisguised competition.

10 The Sparsity and Plurality of Selection Criteria

This section treats the question the staged design raises and does not answer: how selection can occur without becoming the whole meaning of the arrangement. Its role is to convert a familiar warning into two parameters that a community can set and a reader can check. Its structure treats the reactivity problem, the two parameters, their interaction, the costs of setting them, and the limits of the treatment. Its method is analysis with each claim marked, and the section proposes no mechanism the protocol enforces.

10.1 The Reactivity Problem

Where a measure is used to allocate, the measure becomes an object of effort and the categories it recognises become the categories the field maintains (Espeland and Sauder, 2007). The result is not a defect of any particular measure and does not depend on the measure being poorly chosen.

Claim 10.1. *Any selection criterion applied often enough and consistently enough becomes the description under which participants organise their work. The problem is therefore not the choice of criterion but the frequency and singularity of its application. This claim is empirical as to the mechanism, resting on studies of ranking in professional fields, and formal as to the inference.*

10.2 The Two Parameters

Definition 10.2. (Sparsity and plurality). *The sparsity of a selection is the interval between selection events measured against the interval over which a line of work develops. The plurality of a selection is the number of criteria applied that cannot be combined into a single ordering.*

Claim 10.3. *A selection that is sparse in time exerts weak optimisation pressure between events, since a participant cannot organise work continuously around an event that occurs rarely. A selection that is plural in criterion prevents a single ordering from forming, since parties optimising for one criterion are not thereby advantaged on another. A selection that is frequent and single-criterion is the condition under which reactivity is strongest. This claim is formal.*

The two parameters correspond to the two things a community actually controls. It cannot avoid selecting, and it can decide how often and on how many incommensurable grounds.

10.3 Interaction of the Parameters

The parameters are not independent in their effects, and two combinations deserve naming.

Sparse and plural is the condition this section recommends and is expensive: rare selections require larger commitments at each event, and plural criteria require assessors

competent on several dimensions and a decision procedure that does not silently reduce them to one.

Frequent and plural is unstable. Where selections are frequent, participants discover which criterion is in practice decisive and optimise for it, and the plurality becomes nominal. A community that wishes plurality to hold must reduce frequency.

Claim 10.4. *Plurality of criterion is not preserved under frequent application, since participants infer the operative criterion from repeated outcomes. Sparsity is therefore a condition of plurality rather than an independent parameter. This claim is a conjecture, and the mechanism it posits is the ordinary one by which participants learn a system's real rules.*

10.4 The Costs

Three, and the second is the reason communities do not adopt the recommendation.

Sparse selection concentrates risk. A participant who fails at a rare event waits a long time for another, and the arrangement is harsher for those with short horizons, which is again the population with least standing.

Sparse selection is administratively unattractive. Frequent small allocations are easier to justify, easier to correct, and produce a steady record of activity, and the parties who administer allocation have reasons to prefer them that are unrelated to what they do to the field.

Plural criteria are contestable. Where several incommensurable criteria are applied, the decision cannot be defended as following from the criteria, and the decision-maker's judgement becomes visible. This is a cost in the eyes of those who prefer decisions to appear mechanical and is an advantage under this series' analysis, since a visible judgement is attributable to a party.

10.5 The Limits of the Treatment

The protocol enforces neither parameter and can enforce neither, since both concern how a community allocates and the specification takes no position on allocation.

What the record contributes is that selection events are recorded as decisions with their reasons, so a reader can determine how often selections occurred and on what grounds. Whether that determination has any consequence depends on who reads it.

Claim 10.5. *Sparsity and plurality are charter parameters, not protocol parameters. The protocol makes their values visible after the fact and supplies no means of setting them. Recorded under the standard of Section 2, the reactivity difficulty is addressed at the level of communities and unaddressed at the level of the design. This claim is formal.*

10.6 Summary of Results

This section has treated the reactivity of selection criteria. Claim 10.1 establishes that any criterion applied frequently and singly becomes the description under which work is organised, so the problem is frequency and singularity rather than the choice of criterion.

Definition 10.2 fixes sparsity and plurality, and Claim 10.3 states their effects: sparse selection exerts weak pressure between events, plural criteria prevent a single ordering, and frequent single-criterion selection is the worst case. Claim 10.4 states that plurality is not preserved under frequent application, so sparsity is a condition of plurality. Three costs are recorded: sparse selection concentrates risk on those with short horizons, is administratively unattractive, and plural criteria expose the decision-maker's judgement, which is a cost by ordinary standards and an advantage under this series' analysis. Claim 10.5 records that both parameters are charter matters, that the protocol makes their values visible after the fact, and that the difficulty is unaddressed at the level of the design.

11 Private Benefit, Overhead, and the Distribution of Adoption Cost

This section treats the cost the arrangement imposes on those who use it. Its role is to establish who bears that cost, to state the condition an arrangement must satisfy if it is to be adopted at all, and to record what the design does about the distribution. Its structure treats the overhead, its incidence, the byproduct principle, the private-benefit condition, and the residual regressivity. Its method is analysis with each claim marked.

11.1 The Overhead

Conducting work under the protocol requires acts that would not otherwise be performed: typing a transition, referencing the state it alters, registering another party's transition, recording a decision with its reason.

Two of these are cheap by construction. Registration is a single affirmative action, and typing is a small number of narrow choices under the factorisation the specification fixes.

One is not cheap and is the act on which the arrangement's distinctive benefit depends. Recording a decision to abandon a direction, with the reason, at the moment of abandoning it, requires the participant to articulate something they are in the act of setting aside, and articulation is the expensive part.

Claim 11.1. *The overhead of the arrangement is concentrated in the acts requiring articulation, and specifically in the decision act. The mechanical acts are cheap and the interpretive act is not, and no property of an interface reduces the cost of saying why an approach failed. This claim is formal.*

11.2 The Incidence

Overhead is not distributed evenly among participants, and the direction of the unevenness is unfavourable to this series' own position.

A senior party with assistance can absorb the cost of maintaining a record. A doctoral student, a precariously employed researcher, or an unaffiliated scholar cannot, and each additional act competes directly with the output on which their position depends.

Claim 11.2. *The overhead of the arrangement falls hardest on participants with the least time and the least standing, who are the population the series' diagnosis concerns. An arrangement that imposes uncompensated effort on that population is regressive in the dimension the series most cares about, whatever its intentions. This claim is formal, given the distribution of assistance and time in the present system.*

The claim should be read as a constraint rather than as an objection defeated elsewhere. It is not answered by the benefits the arrangement supplies, since those benefits are also unevenly distributed and arrive later.

11.3 The Byproduct Principle

The design's response is to require that every act be one a participant performs for a reason of their own.

The principle is satisfied for most of the vocabulary. An objection is raised because the objector wants a claim changed. A release is declared because the author wants something citable. A connection is recorded because the connector wants the relation known. In each case the record falls out of an act performed for its own sake.

The strongest form of the principle appears where the arrangement records what participants already produce. Weekly progress reports, supervision notes and meeting summaries are compressed accounts of a trajectory, produced routinely under present conditions and discarded. An arrangement that adds a small amount of structure to work already performed imposes far less than one that asks for new work.

Claim 11.3. *The byproduct principle reduces the overhead of every act except the decision act, whose purpose for the performer is weakest. A participant abandoning a direction gains little from recording why, and the party who gains is a stranger years later. This is the disparity that defeated earlier systems, surviving in one act after the design has removed it from the others. This claim is formal.*

11.4 The Private-Benefit Condition

Requirement 11.4. (Private-benefit sufficiency). *An arrangement is adoptable only where its minimal form is worth adopting by a single participant with no other participants present. An arrangement whose benefits appear only at scale is never adopted, since no party is ever in a position to obtain them.*

The condition is satisfied by the specification's personal tier, and the benefits it supplies are modest and real: a searchable record of what was ruled out and why, a standing list of unanswered objections, and a release that emits a citable document with its chain.

The historical analogue is worth stating because it is the only kind of evidence available. Version control was adopted by programmers working alone, for private reasons, and the collaborative benefits followed. Nobody adopted it out of commitment to a social model of software production.

Claim 11.5. *Adoption of an arrangement of this kind proceeds from private utility to collective benefit and not in the reverse direction. Any argument for adoption that appeals to what the arrangement would achieve at scale is an argument no individual can act on. This claim is a conjecture supported by analogy, and the analogy is to a case with far more favourable cost structure.*

11.5 The Residual Regressivity

The private-benefit condition addresses the timing of benefits and not their distribution.

A senior party adopting the personal tier obtains a record and loses little. A precarious party adopting it obtains the same record and loses time they cannot spare. The condition ensures that both obtain something; it does not equalise what they give up.

Claim 11.6. *The distribution of adoption cost remains regressive under the byproduct principle and the private-benefit condition, and the design contains no mechanism that alters it. Recorded under the standard of Section 2, this is an unaddressed difficulty: a mechanism that compensated the cost would require a party to fund it, and no such party is identified. This claim is formal.*

11.6 Summary of Results

This section has treated adoption cost. The overhead concentrates in acts requiring articulation, and the decision act is the expensive one. Its incidence falls hardest on participants with least time and standing, which is the population the series concerns, so the arrangement is regressive in the dimension it most cares about. The byproduct principle reduces the cost of every act whose purpose is the performer's own, and its strongest form records what participants already produce and discard; the decision act is where the principle is weakest, and the disparity that defeated earlier systems survives there. Requirement 11.4 fixes private-benefit sufficiency, satisfied by the personal tier, with version control as the historical analogue and with the analogy's more favourable cost structure noted. The residual regressivity is recorded as unaddressed, since compensating it would require a funding party that no analysis identifies.

12 The Selection Pressure on Adopting Groups

This section treats the environment in which a group deciding whether to adopt is situated. Its role is to state the strongest argument against adoption occurring at all, and to identify the only two escapes from it. Its structure treats the environment, the dominance argument, the two escapes, the prediction that follows, and the honesty condition. Its method is analysis, and the section reaches a conclusion unfavourable to the proposal.

12.1 The Environment

A research group survives on outputs. Publications and their reception determine grants, grants determine positions and equipment, and positions determine whether the group

continues. The environment rewards output rate and rewards it in a currency that compounds (Merton, 1968).

12.2 The Dominance Argument

Claim 12.1. *In an environment rewarding output rate, a group adopting an arrangement that imposes any net cost is disadvantaged relative to groups that do not, irrespective of the arrangement's epistemic merits. Where the disadvantage is sustained, adopting groups are selected against. This claim is formal.*

The argument is the reason this paper does not treat adoption as a matter of persuasion. A group persuaded of the arrangement's value and situated in this environment is a group that has been persuaded to accept a competitive disadvantage, and persuasion of that kind does not survive a funding cycle.

The argument also disposes of a common response. That the collective benefits of adoption exceed the collective costs is not in dispute and is not relevant, since no group receives the collective benefit and every adopting group bears its own cost. This is the standard structure of collective-action failure (Olson, 1965).

12.3 The Two Escapes

Net private benefit. Where the arrangement returns more to the adopting party than it costs them, at their own scale and without others, the dominance argument does not apply. This is the private-benefit condition of the previous section, and it is why that condition is not a design preference but the escape route on which adoption depends.

The candidate benefits are stated in the specification's tier table and are worth restating as an economic claim rather than a feature list: time recovered in preparing progress reports and presentations from an existing record, effort saved in not repeating an abandoned approach, and faster onboarding of a new member from the record in place of from conversation. Whether these exceed the overhead is an empirical question nobody has measured.

Insulation. Where a party is not subject to output-rate selection, the dominance argument does not bind. Theory groups with secure funding, nonprofit research bodies, interdisciplinary institutes, retired scholars and unaffiliated researchers are insulated to varying degrees.

Claim 12.2. *Adoption is predicted to begin among parties insulated from output-rate selection, and to spread to parties subject to it only where net private benefit is demonstrated. The prediction is uncomfortable for the series, since insulated parties are peripheral to the fields whose practice the arrangement would have to change. This claim is formal as to the prediction and a conjecture as to its consequence.*

12.4 The Complementary Position

One further escape is available and is weaker than it appears.

An arrangement entering as a complement, which asks nobody to abandon publication and produces the artefact the incumbent system accepts, imposes a smaller net cost and faces less organised opposition. The specification's requirement that a release emit a citable document is the mechanism.

Claim 12.3. *Complementary entry defers the conflict with parties whose position depends on the artefact's evidential function; it does not dissolve it. An arrangement that came to carry that function would displace what those parties supply, and their interest would then be what it is now. This claim is formal.*

The point matters for how the series presents itself. A proposal claiming to threaten nobody while proposing to change what counts as evidence will be read as naive or as evasive, and the honest statement is that the conflict is postponed.

12.5 The Honesty Condition

Claim 12.4. *The series may not argue for adoption on the ground that the arrangement would benefit scholarship collectively. That argument is available to no individual party and its repetition is the characteristic error of proposals in this field. What may be argued is that a particular party, at a particular scale, obtains more than it gives up, and that argument requires evidence this series does not have. This claim is formal.*

12.6 Summary of Results

This section has stated the strongest argument against adoption. A group adopting an arrangement with any net cost is disadvantaged in an environment rewarding output rate, and sustained disadvantage selects adopting groups out; the collective benefit is not relevant, since no group receives it. Two escapes exist: net private benefit at the adopting party's own scale, whose candidate components are time recovered in reporting, effort saved in not repeating abandoned work, and faster onboarding, none measured; and insulation from output-rate selection, which predicts adoption beginning among theory groups, nonprofit bodies, interdisciplinary institutes and unaffiliated scholars, a prediction uncomfortable because those parties are peripheral to the practice the arrangement would change. Complementary entry defers the conflict with incumbents and does not dissolve it. Claim ?? forbids the series from arguing for adoption on collective grounds.

13 Network Externalities, Conformance Tiers, and the Minimum Viable Adopting Unit

This section treats the coordination problem that arises where a good becomes more valuable as others use it. Its role is to show how the tier structure of the specification addresses the problem, and to state what it leaves standing. Its structure treats the externality, the ordinary remedies and why they are unavailable here, the tier structure as a response, the minimum viable adopting unit, and the residual difficulty. Its method is analysis with each claim marked.

13.1 The Externality

Several of the arrangement's benefits rise with the number of participants. Attestation requires a second party. Absorption requires a line of work to absorb from. Structural matching across fields requires a corpus. Propagation requires a second implementation.

Claim 13.1. *An arrangement whose principal benefits rise with participation faces a coordination problem: each party's return depends on decisions taken by others, and the return to the first adopter is lowest. Where the benefit is entirely of this kind, the arrangement has no first adopter and does not begin. This claim is formal.*

The claim explains a pattern the companion paper documents. Representations of scholarly reasoning have existed for decades and have not been adopted, and the accounts given have concentrated on capture cost. The coordination problem is a second and independent reason: a party constructing an argument map alone obtains a map, and the value the systems promised lay in a network that never formed.

13.2 The Ordinary Remedies and Their Unavailability

Three remedies are standard and each is unavailable here.

Subsidy of early adopters requires a funding party with an interest proportional to the eventual benefit, which the analysis of maintainer labour has already established does not exist.

Mandate requires an authority able to compel participation. Funders and institutions can compel, and the arrangement would then be adopted as a compliance requirement, which the companion papers' analysis of capture treats as the condition under which the arrangement's purposes become the mandating party's purposes.

Bundling with an already adopted good requires a good that participants already use and into which the arrangement can be placed. This is the most promising of the three, and the specification's decision to layer over version control and repositories is an attempt at it, whose success is unknown.

13.3 The Tier Structure as a Response

The specification's response is structural rather than promotional.

Claim 13.2. *Where an arrangement's benefits are stratified so that each tier returns something at the scale of that tier, the coordination problem is decomposed into a sequence of decisions each of which a party can take alone. The personal tier returns a record to one participant; the group tier returns attestation and attribution to a group; the open tier returns propagation and entry to a community. No party is asked to act on a benefit that requires others. This claim is formal.*

The condition the structure must satisfy is strict and is worth stating as such: each tier must return something at its own scale, and a tier that returns nothing until the next is reached collapses the decomposition. The personal tier is where this is most doubtful, and the previous sections have already recorded that the benefits available there are

modest.

13.4 The Minimum Viable Adopting Unit

Definition 13.3. (Minimum viable adopting unit). *The minimum viable adopting unit is the smallest set of parties for which the arrangement's benefits exceed its costs without participation by anyone outside the set.*

Claim 13.4. *Under the specification's tier structure the minimum viable adopting unit at the group tier is two parties who register each other's transitions, and at the open tier is one community with an implementation and a charter. The unit is small enough that adoption does not require coordination across a field. Whether the benefits at those units exceed the costs is an empirical question, and this claim concerns only the size of the unit. This claim is formal.*

The smallness matters because it determines what a first adopter must arrange. A design whose minimum unit was a discipline would require an agreement nobody can broker. A design whose minimum unit is a seminar requires a decision two people can take on a Tuesday.

13.5 The Residual Difficulty

Two items remain and the second is the important one.

The cross-field matching benefit is not decomposable. It requires a corpus containing trajectories from several fields, so it belongs to no tier and returns nothing to any adopting party. It should therefore play no part in any argument for adoption, and the companion papers restrict it accordingly.

And the tier structure decomposes the coordination problem without addressing the cost problem. A party for whom the personal tier returns less than it costs does not adopt, however small the unit, and the previous sections have recorded that this describes exactly the participants with least time.

Claim 13.5. *The tier structure addresses the coordination problem and does not address the cost problem, and the two are independent. An arrangement may have a small minimum viable unit and still not be adopted, if the cost at that unit exceeds the return. This claim is formal.*

13.6 Summary of Results

This section has treated the coordination problem. Claim 13.1 states that an arrangement whose benefits rise with participation has its lowest return at the first adopter, and that where the benefit is entirely of this kind the arrangement does not begin, which supplies a second and independent explanation for the non-adoption of earlier systems. The three ordinary remedies are unavailable: subsidy lacks a funding party, mandate converts the arrangement into a compliance requirement and invites capture, and bundling

is what the layering over existing substrates attempts. Claim 13.2 states the tier structure's response, decomposing the coordination problem into decisions a party can take alone, under the strict condition that each tier return something at its own scale. Definition 13.3 fixes the minimum viable adopting unit, and Claim 13.4 places it at two parties for the group tier and one community for the open tier. Two items remain: cross-field matching is not decomposable and must play no part in adoption arguments, and the tier structure leaves the cost problem untouched.

14 Compatibility with the Incumbent Reward System

This section treats what an adopting party must give up. Its role is to establish that the arrangement requires no sacrifice of the currency in which scholarly careers are conducted, and to state the limits of that compatibility. Its structure treats the requirement, the mechanism, what compatibility secures, the objection that compatibility concedes too much, and the point at which it ends. Its method is analysis, with the strategic consequences stated.

14.1 The Requirement

Requirement 14.1. (Compatibility). *An arrangement is adoptable only where a participant can obtain, from a conforming record and without additional work, an artefact of the kind the incumbent reward system accepts.*

The requirement follows from the analysis of the two preceding sections. A participant subject to output-rate selection who adopts an arrangement that does not produce publications has accepted a cost with no offsetting return in the currency that determines their position, and the dominance argument applies with full force.

14.2 The Mechanism

The specification requires that a release emit a document containing the released state, the chain of transitions leading to it, the parties with their contributor roles, the absorption links with their attributions, and the objections standing at release.

Two properties of this emission matter economically.

It requires no additional work. The document is assembled from records created in the course of the work, so the participant obtains the artefact as a byproduct of the arrangement rather than as a separate undertaking.

And it carries an appendix no other process can produce. A publication accompanied by the chain that produced it, with contributors attributed to changes and standing objections enumerated, is a document with properties the incumbent system does not presently receive, whatever it does with them.

Claim 14.2. *Compatibility converts the adoption decision from a choice between two systems into an addition to one. A participant continues to publish and obtains the publication with less effort than before, and the question of whether the record is valued arises later and separately. This claim is formal.*

14.3 The Effects of Compatibility

Three things, of which the third is strategic.

It removes the sacrifice. Nothing in the arrangement asks a participant to forgo publication, citation or the ordinary conduct of a career.

It supplies a return in the currency that binds. Time recovered in assembling a paper from an existing record is a return a participant receives immediately and in the dimension the environment rewards.

And it defers organised opposition. Parties whose position depends on the artefact's evidential function have no immediate reason to oppose an arrangement that produces artefacts, and the analysis of concentrated interests against diffuse benefits (Olson, 1965) predicts that opposition organises when a threat becomes visible.

14.4 The Objection That Compatibility Concedes Too Much

An objection is available and deserves an answer, since it is the objection a sympathetic reader will raise.

The objection holds that an arrangement designed to feed the incumbent system entrenches it: participants continue to publish, publications continue to be counted, and the trajectory record becomes an unread appendix.

The answer is partial. The objection is correct that compatibility entrenches nothing about the record's status, and correct that the appendix may go unread indefinitely. What it misses is the alternative: an arrangement demanding that participants abandon publication asks them to accept the dominance argument's cost in full, and would be adopted by nobody, so the comparison is between an arrangement that may be ignored and one that will not exist.

Claim 14.3. *Compatibility risks the record becoming an unread appendix and is the condition under which any record exists at all. The choice is between an arrangement whose evidential value may never be recognised and an arrangement that is not adopted. This claim is formal, and the judgement that the first is preferable is a position rather than a result.*

14.5 The Terminus of Compatibility

The deferral has a terminus and the series should name it.

An arrangement whose records came to be read as evidence of capacity would displace part of what the artefact presently supplies. Parties whose position rests on that supply, including those who assess by counting and those who publish for a living, would then have the interest they do not have now, and the political economy of the preceding section would apply in full.

Claim 14.4. *Compatibility postpones the conflict with incumbent interests and does not remove it, and the conflict returns at exactly the point at which the arrangement begins to succeed. A proposal claiming to threaten nobody while proposing to change what counts as evidence is either naive or evasive, and this series states the terminus rather than claiming the*

conflict away. This claim is formal.

14.6 Summary of Results

This section has treated compatibility. Requirement 14.1 fixes it as a condition of adoptability, following from the dominance argument. The mechanism is emission from a release, which requires no additional work and carries an appendix the incumbent system does not presently receive. Compatibility removes the sacrifice, supplies an immediate return in the binding currency, and defers organised opposition. The objection that compatibility entrenches the incumbent system is partly correct and is answered by the alternative, since an arrangement demanding the abandonment of publication would not be adopted. Claim 14.4 states the trade plainly, between a record whose evidential value may never be recognised and no record at all. Claim ?? names the terminus: the conflict returns at the point at which the arrangement begins to succeed.

15 The Empirical Record of Research Infrastructure Adoption

This section examines what is known about why research infrastructures are adopted or abandoned. Its role is to test the account of the preceding sections against evidence, and to state honestly how weak that evidence is. Its structure treats the character of the evidence, the pattern the cases suggest, the funded and mandated case, the counter-evidence on encounter, and what the record does and does not support. Its method is a reading of cases with their selection bias stated first, and every claim in the section is marked at the strength the evidence permits.

15.1 The Character of the Evidence

Claim 15.1. *No systematic study exists of a defined population of research infrastructures assessed against adoption outcomes. The available evidence consists of retrospective accounts of individual systems, written mostly by parties associated with them, and selected for attention because they succeeded or failed conspicuously. Inference from it is accordingly weak. This claim is empirical and concerns the state of a literature.*

The selection problem is severe and runs in the direction that flatters the account offered here. Infrastructures that were quietly built and quietly unused generate no literature at all, and the population against which any pattern should be assessed is therefore unobservable in exactly the region where the account makes its prediction.

15.2 The Pattern the Cases Suggest

With that stated, the cases that are discussed exhibit a pattern consistent with the analysis of Sections 11 and 12.

Infrastructures that supplied a tool researchers needed for their own work, and in which sharing was a byproduct of using the tool, were taken up. Simulation and analysis platforms in engineering and in the life sciences are the cases usually named, and in each the user obtained a capability at the moment of use.

Infrastructures whose proposition was sharing, in which the contribution was the point and the return arrived through others, were built and largely unused. Workflow-sharing repositories are the case usually named, and the accounts of their decline describe exactly the disparity of Section 11.

Claim 15.2. *The cases discussed in the literature are consistent with the private-benefit condition: infrastructures returning something at the moment of use were adopted and infrastructures returning something through the participation of others were not. The consistency is suggestive and is not evidence, given the selection problem stated above. This claim is a conjecture.*

15.3 The Funded and Mandated Case

One class of case bears directly on the remedies dismissed in Section 13 and deserves separate treatment.

Large collaborative research programmes have built shared infrastructure with substantial funding and with institutional expectation that participants use it. The accounts of such programmes describe infrastructure that was built and maintained, and participation in its collaborative components that fell well short of what the investment implied, alongside sustained scientific criticism of the programmes themselves.

Claim 15.3. *Funding and institutional expectation are sufficient to build shared research infrastructure and are not sufficient to produce voluntary use of it. The proposition is consistent with the analysis of Section 12, since a mandate alters what a participant must be seen to do and does not alter the cost the participant bears. This claim is a conjecture, resting on retrospective accounts of a small number of large programmes.*

15.4 The Counter-Evidence on Encounter

One body of evidence runs against a claim the series makes, and it is stated here rather than in the paper that makes the claim.

The companion paper motivates the arrangement partly by the value of unplanned encounter. The best qualitative evidence available concerns physical rather than digital arrangements and is unfavourable: a study of a building designed to produce collaboration through proximity found that its occupants developed strategies that minimised serendipitous encounters and avoided new collaborations (Irving et al., 2020). Work relating spatial layout to tie formation reports associations without settling their direction (Kabo et al., 2014).

Claim 15.4. *There is no evidence that an infrastructure increases productive unplanned encounters, and the strongest qualitative study of an arrangement designed to produce them reports avoidance. Claims in this series about encounter must accordingly be confined to the cost of preparation and must never assert an increase in frequency. This claim is empirical as to the finding and formal as to the restriction it imposes.*

15.5 The Statements the Record Supports

Three statements, at the strength the evidence permits.

That an infrastructure returning nothing at the moment of use is unlikely to be adopted is consistent with every case discussed and is supported by no systematic study.

That funding and mandate build infrastructure without producing use is consistent with the large-programme accounts and rests on a small number of cases.

That an arrangement of the kind this series proposes would be adopted is supported by nothing. No instance exists, and the analogy to version control concerns a case with a far more favourable cost structure and a far larger population of dependent users.

Claim 15.5. *The empirical record supports the negative claims of this paper and supports none of its positive ones. It gives reason to expect that arrangements without private benefit fail, and gives no reason to expect that an arrangement with private benefit succeeds. This claim is formal, given the state of the evidence described above.*

15.6 Summary of Results

This section has examined the evidence and found it weak. No systematic study of a defined population exists, the available accounts are retrospective and selected, and quietly unused systems generate no literature, so the population is unobservable exactly where the account predicts. The cases discussed are consistent with the private-benefit condition, with tool-supplying infrastructures adopted and sharing-proposition infrastructures unused, and the consistency is suggestive rather than evidential. Funded and mandated programmes build infrastructure without producing voluntary use, which is consistent with the dominance argument. The evidence on encounter runs against the series' motivation and restricts its claims to the cost of preparation. The record supports this paper's negative claims and none of its positive ones.

16 Predictions and the Conditions of Falsification

This section states what the account predicts and what would show it wrong. Its role is to make the analysis testable, since an account of incentives that predicts nothing can be neither used nor refuted. Its structure treats the predictions in three groups, the observations that would refute each, the measurement difficulties, and the prediction on which the account is most exposed. Its method is enumeration, with each prediction derived from a named claim above.

16.1 Predictions Concerning Adoption

P1. Adoption begins among parties insulated from output-rate selection, and among parties for whom the reporting burden is high relative to their output. Derived from Claim 12.3.

P2. Adoption proceeds tier by tier, with a substantial population at the personal tier preceding any population at the group tier. Derived from Claim 13.2.

P3. Arrangements adopted for a private benefit persist and arrangements adopted for a collective one do not, at the level of the individual group. Derived from Requirement 11.4 and the record of Section 15.

16.2 Predictions Concerning Behaviour within an Arrangement

P4. Where selection is present and absorbed content is not attributed, objection is disproportionately destructive and disclosure of strong material is late. Derived from Claim 9.2.

P5. Where absorbed content is attributed and the attribution is recognised, contribution to lines of work other than one's own rises. Derived from Claim 9.3.

P6. Where selection is frequent, plurality of criteria decays and participants converge on the criterion that is in practice decisive. Derived from Claim 10.4.

P7. Assessment capacity, not contribution volume, becomes the binding constraint on any arrangement that succeeds. Derived from Claim 4.1.

16.3 Predictions Concerning the Distribution of Effects

P8. The overhead of participation falls disproportionately on participants with least time, and participation rates by career stage reflect it. Derived from Claim 11.2.

P9. Maintenance concentrates on a small number of parties whose number does not grow with adoption. Derived from Claim 5.2.

P10. Attribution mechanisms are resisted by parties who benefit from the present invisibility of others' contributions, and the resistance appears as objection to the mechanisms' accuracy in place of objection to their purpose. Derived from Claim 6.2.

16.4 The Refuting Observations

Each prediction has an observation that would refute it, and they are stated together so that a reader may look for them.

Adoption concentrated among parties under strong output-rate selection, without demonstrated private benefit, refutes *P1* and the dominance argument with it.

Group-tier populations arising without a preceding personal-tier population refutes *P2* and the decomposition argument.

Persistence of arrangements adopted on collective grounds refutes *P3* and Requirement 11.4.

Constructive objection in arrangements with unattributed selection refutes *P4* and, with it, the backward-induction analysis.

Stable plurality of criteria under frequent selection refutes *P6*.

Contribution volume rather than assessment capacity binding in a successful arrangement refutes *P7* and the location of the commons dilemma in Section 3.

Participation rates independent of career stage refute *P8*.

Maintenance distributed broadly and stably refutes *P9* and the reading of the software record.

16.5 The Measurement Difficulties

Four, each stated against the account's own interest.

Non-adoption is unobservable. Parties who consider the arrangement and decline leave no record, so P1 and P3 are testable only among adopters, which is the population selected for the outcome the predictions concern.

Attribution recognition is unobservable directly. P5 depends on whether communities count absorbed contributions, and the counting occurs in appointment and funding decisions that are not public.

Destructive objection is hard to distinguish from rigorous objection. P4 concerns the intent behind an act whose surface is identical either way, and any test would rest on judgements of a kind this series has excluded from the protocol.

And every prediction concerns an arrangement that does not exist. The account is presently untested in the only population that could test it.

16.6 The Prediction on Which the Account Is Most Exposed

Claim 16.1. *The account is most exposed on P2. If a substantial personal-tier population appears and no group-tier population follows, the tier decomposition will have delivered private utility without the attested records on which every evidential claim in the series depends, and the arrangement will have become a note-taking convention. This outcome is consistent with everything the account asserts and would defeat the purpose the series was written for. This claim is formal.*

16.7 Summary of Results

This section has stated ten predictions with their derivations: three concerning adoption, four concerning behaviour within an arrangement, and three concerning the distribution of effects. Refuting observations are given for eight of them, and four measurement difficulties are recorded, of which the first, that non-adoption is unobservable, bears on the two adoption predictions directly. Claim 16.1 identifies P2 as the prediction on which the account is most exposed, since a personal-tier population without a group-tier successor would satisfy the account and defeat the purpose.

17 The Boundaries of the Account and Its Conclusion

This section states what the paper has and has not shown. Its role is to close with a summary a reader can hold against the argument, and to record the difficulties left unaddressed in one place. Its structure treats the results, the unaddressed difficulties, what the paper declines to argue, and the conclusion. Its method is enumeration.

17.1 The Results

Five.

The commons dilemma in an arrangement of this kind attaches to review capacity,

maintainer labour, the instrument and community trust, and not to knowledge, whose failure mode is underprovision. Remedies of the appropriation kind treat a problem that does not obtain.

Recognition is positional, so the aggregate available is fixed and attribution mechanisms redistribute rather than create it. No claim that the arrangement reduces competition may stand anywhere in the series, and the claims about retention, reuse and interpretability are unaffected by this bound.

An open selection is a contest whose aggregate effort is dissipated, and what distinguishes a generative arrangement from a competitive one is whether content from unselected work enters what proceeds with attribution. The test is checkable from the record, applies to arrangements without one, and must never be counted.

Attributive absorption is the condition under which a staged design survives backward induction. Without it, such a design predicts weak early disclosure and destructive objection, and is worse than an undisguised competition.

And adoption depends on private benefit at the adopting party's own scale, since a group adopting an arrangement with any net cost is selected against in an environment rewarding output rate. Adoption is predicted to begin among insulated parties, which is uncomfortable for the series.

17.2 The Unaddressed Difficulties

Under the standard fixed in Section 2, four difficulties are recorded as unaddressed, meaning that no mechanism has been stated with an identified party for whom bearing its cost is rational.

Congestion of assessment capacity. The remedies available all place an unfunded burden on a few parties, and the exclusion of scalar measures removes the remedy in general use.

Maintainer labour. Unfunded, concentrating, and growing with adoption, with no party whose interest is proportional to the benefit.

The regressive distribution of overhead. The byproduct principle and the private-benefit condition address the timing and the magnitude of costs and not their incidence.

And the recognition of attributed contribution, on which the repair of staged selection depends and which no property of a protocol supplies.

17.3 The Arguments Declined

Three refusals, each of which removes a move the literature makes routinely.

It does not argue that the present arrangement is defective enough that any alternative is worth trying. The present arrangement performs two functions the account must preserve, and a proposal degrading either without replacement leaves the system worse.

It does not argue for adoption on collective grounds. That argument is available to no individual party and its repetition is the characteristic error of this field.

And it does not claim that the empirical record supports the proposal. The record supports the paper's negative claims, gives reason to expect that arrangements without

private benefit fail, and gives no reason to expect that an arrangement with private benefit succeeds.

17.4 Conclusion

The difficulties this paper treats are not technical and are not removed by the specification of a protocol. They are the ordinary difficulties of collective action, applied to a case in which the good produced is non-rival, the goods consumed are not, and the currency in which participants are paid is positional and fixed.

What the analysis yields is narrow and, so far as it goes, usable. It locates the dilemma where it actually sits, so that governance effort is not spent restricting the use of knowledge. It supplies one test that distinguishes an arrangement that generates from one that selects, and the test is checkable and falsifiable. It identifies the single condition under which a staged design does not collapse under strategic behaviour. And it states the condition on which adoption depends, in a form that permits a particular party to assess it for themselves.

What it does not yield is a reason to expect that any of this occurs. The account is compatible with an outcome in which the arrangement is adopted by a small number of insulated parties, delivers private utility, never reaches the scale at which its evidential claims mean anything, and is remembered as one more proposal of a kind that has been made since 1970. That outcome is stated here because the account predicts it, and because a paper that concealed it would exhibit the failure it spends its length diagnosing.

17.5 Summary of Results

This section has stated five results: the location of the commons dilemma, the positional bound on recognition and the claims it forbids, the dissipation of open selection with the absorption test as the distinguishing criterion, attributive absorption as the condition on staged selection, and private benefit as the condition on adoption. Four difficulties are recorded as unaddressed: assessment congestion, maintainer labour, the regressive incidence of overhead, and the recognition of attributed contribution. Three arguments are declined: that the present arrangement's defects justify any alternative, that adoption follows from collective benefit, and that the empirical record supports the proposal. The conclusion states what the analysis yields and states that it yields no reason to expect adoption.

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